SUKANTO BISWAS

M.Sc. Student, Materials and Metallurgical Engineering, BUET

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RESEARCH PROFILE

M.Sc. student in Materials and Metallurgical Engineering at BUET with a B.Sc. in Mechanical Engineering. Three peer-reviewed publications in heat transfer, CFD and aerodynamics. Currently researching cooling of bifacial PV modules, and seeking a Master's/PhD position on next-generation photovoltaic devices.

Research interests: Perovskite and bifacial photovoltaics · Thermal management of solar cells · Heat transfer and fluid flow · Computational fluid dynamics (CFD)

EDUCATION

M.Sc. in Materials and Metallurgical Engineering Ongoing

Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh

- Research focus: perovskite solar cells

B.Sc. in Mechanical Engineering 2023

Khulna University of Engineering and Technology (KUET), Khulna, Bangladesh

- CGPA: 2.85/4.00
- Thesis: *Heat Transfer and Fluid Flow Analysis in a Corrugated Plate Heat Exchanger*

RESEARCH EXPERIENCE

Cooling System Design for Bifacial Solar Cells Present

Independent research

- Investigating thermal regulation of bifacial PV modules, where rear-side irradiance raises operating temperature and cuts conversion efficiency.
- Building heat-transfer models of candidate cooling mechanisms to quantify efficiency gains.
- Designing combined numerical and experimental studies for device-level optimization.

Heat Transfer and Fluid Flow in a Corrugated Plate Heat Exchanger 2022

Undergraduate research, KUET

- Ran CFD simulations in ANSYS Fluent of flow and heat transfer in corrugated channels.
- Validated numerical results against theoretical correlations.
- Presented at ICMIEE-2022; published in the conference proceedings (2025).

Collaborative CFD Studies 2025

- Numerical study of fin geometry and material for air-cooled engine cylinders (*Next Research*, 2025).
- Compressible-flow study of Mach number effects on a biconvex airfoil (*Engineering and Applied Science*, 2025).

PUBLICATIONS

Journal Articles

1. S. Bakchi, R. Mustak, D. Biswas, and **S. Biswas**, "Numerical investigation on optimizing engine cooling using different fin geometry and material on an engine cylinder," *Next Research*, vol. 2, no. 3, Art. no. 100674, Sep. 2025. doi:10.1016/j.nexres.2025.100674

2. D. Biswas, R. Mustak, S. Bakchi, and **S. Biswas**, "Influence of Mach number on the aerodynamic performance of a biconvex airfoil," *Engineering and Applied Science*, vol. 10, no. 2, pp. 11–17, Apr. 2025. doi:10.11648/j.eas.20251002.11

Conference Proceedings

3. **S. Biswas**, M. I. Inam, and P. C. Roy, "Heat transfer and fluid flow analysis in a corrugated plate heat exchanger," in *Proc. 7th Int. Conf. on Mechanical, Industrial and Energy Engineering (ICMIEE-2022)*, Khulna, Bangladesh, Dec. 22–24, 2022; *SciEn Conference Series: Engineering*, vol. 1, pp. 107–112, 2025. doi:10.38032/scse.2025.1.20

TECHNICAL SKILLS

Simulation	ANSYS Fluent (CFD for heat transfer and compressible flow)
CAD	SOLIDWORKS, AutoCAD
Programming	C/C++, Arduino
Data analysis	OriginPro, Microsoft Excel
Graphics	Adobe Illustrator, Adobe Photoshop
Languages	Bengali (native), English (professional working proficiency)

LEADERSHIP AND AWARDS

- **Founding Team Member**, KUET Mars Rover — Team Durbar
- **Project Manager and Instructor**, LOOP (Control Engineering Club), KUET
- **Class Representative**, Mechanical Engineering Association, KUET
- **2nd Place**, Project Making Competition, Programming and Introduction to Robotics Workshop, LOOP, KUET, 2018
- **2nd Place**, Macroinnovators, IEEE SS12, IEEE Student Branch, KUET, 2018
- **3rd Place**, Technical Drawing and CAD Competition, CADers, KUET, 2017
- **Participant**, "GPH Ispat – Esho Robot Banai" 72-hour robotics hackathon, GPH Group

REFERENCES

Available upon request.